

PayFlow GST - Database Design Document

Database: PostgreSQL

Backend: Node.js (Express)

ORM: Prisma

Architecture: Multi-tenant SaaS

This document defines entities, relationships, constraints, indexes, and database structure for the PayFlow GST application.

users

Columns:

- id UUID PK
- business_id UUID FK
- name VARCHAR(150)
- email VARCHAR(255) UNIQUE
- phone VARCHAR(20)
- password_hash TEXT
- role ENUM(owner,manager,staff)
- is_active BOOLEAN
- created_at TIMESTAMP
- updated_at TIMESTAMP

businesses

Columns:

- id UUID PK
- name VARCHAR(255)
- gstin VARCHAR(15) UNIQUE
- address TEXT
- phone VARCHAR(20)
- email VARCHAR(255)
- logo_url TEXT
- created_at TIMESTAMP

customers

Columns:

- id UUID PK
- business_id UUID FK
- name VARCHAR(255)
- phone VARCHAR(20)
- email VARCHAR(255)
- gstin VARCHAR(15)
- address TEXT
- created_at TIMESTAMP

invoices

Columns:

- id UUID PK
- business_id UUID FK
- customer_id UUID FK
- invoice_number VARCHAR(50) UNIQUE
- invoice_date DATE
- due_date DATE
- subtotal DECIMAL(12,2)
- gst_amount DECIMAL(12,2)
- discount_amount DECIMAL(12,2)
- total_amount DECIMAL(12,2)
- status ENUM(draft,unpaid,partial,paid,overdue)
- created_at TIMESTAMP

invoice_items

Columns:

- id UUID PK
- invoice_id UUID FK
- product_name VARCHAR(255)
- quantity DECIMAL(10,2)
- unit_price DECIMAL(12,2)
- gst_rate DECIMAL(5,2)
- line_total DECIMAL(12,2)

payments

Columns:

- id UUID PK
- invoice_id UUID FK
- amount DECIMAL(12,2)
- payment_method ENUM(cash,upi,bank,cheque)
- reference_no VARCHAR(100)
- payment_date DATE
- created_at TIMESTAMP

reminders

Columns:

- id UUID PK
- invoice_id UUID FK
- channel ENUM(email,whatsapp)
- status ENUM(pending,sent,failed)
- scheduled_at TIMESTAMP
- sent_at TIMESTAMP

audit_logs

Columns:

- id UUID PK
- user_id UUID FK
- entity_type VARCHAR(50)
- entity_id UUID
- action VARCHAR(100)
- metadata JSONB
- created_at TIMESTAMP

Entity Relationships

Business 1:N Users

Business 1:N Customers

Business 1:N Invoices

Customer 1:N Invoices

Invoice 1:N InvoiceItems

Invoice 1:N Payments

Invoice 1:N Reminders

User 1:N AuditLogs

Indexes

- users(email) UNIQUE
- businesses(gstin) UNIQUE
- invoices(invoice_number) UNIQUE
- invoices(customer_id)
- invoices(status)
- payments(invoice_id)
- customers(business_id,name)

Business Rules

- Invoice number must be unique.
- Payments cannot exceed invoice total.
- Outstanding amount = invoice total - payments received.
- Paid invoice status updates automatically.
- Overdue invoices calculated using due date.
- Users can access only their business data.

Future Tables (V2)

- products
- inventory
- purchase_orders
- warehouses
- gst_filings